PORTFOLIO CASE STUDY

Telco Customer Churn

Executive Summary & Technical Architecture

End-to-end Data Engineering & Analytics | Python · MySQL · Advanced SQL · Power BI

Star Schema	12,500+	Months 7–11	Tier 2
1 dimension + 2 fact tables	raw ticket events condensed to scorecards	pre-churn behavioral window	premium segment under investigation

EXECUTIVE SUMMARY

Standard churn analyses stop at a flat file and a model score. This project instead reproduces the **data landscape of a real enterprise**: a flat, Kaggle-style telecom dataset was reverse-engineered into a relational warehouse with event-level operational data, then interrogated with a production-style SQL analytics engine. The objective was to test a specific, falsifiable behavioral hypothesis about premium customers and deliver a clean, BI-ready dataset that isolates the at-risk population.

CORE HYPOTHESIS

Do premium ('Tier 2') customers exhibit a distinct behavioral anomaly, a high volume of 'Escalated' support tickets, between their 7th and 11th month of subscription, just before churning?

SOLUTION ARCHITECTURE

PHASE 1
ETL & Data Generation
Python · pandas · numpy

- Reverse-engineered a flat CSV into normalized relational entities
- Generated synthetic event-based data to emulate a live operational environment
- Enforced primary-key integrity with `drop_duplicates`
- Exported clean CSVs for bulk ingestion

PHASE 2
Database Modeling
MySQL · Star Schema

- Designed one dimension and two fact tables (detailed below)
- Resolved `secure-file-priv` (Error 1290) on a locked-down server
- Executed high-volume `LOAD DATA INFILE` bulk ingestion

PHASE 3
Analytical Engine
Advanced SQL

- Built a single Master Query joining all tables with zero row duplication
- CTE-driven design, window functions and conditional aggregation
- Output: one scorecard row per customer, ready for Power BI

DATA MODEL (STAR SCHEMA)

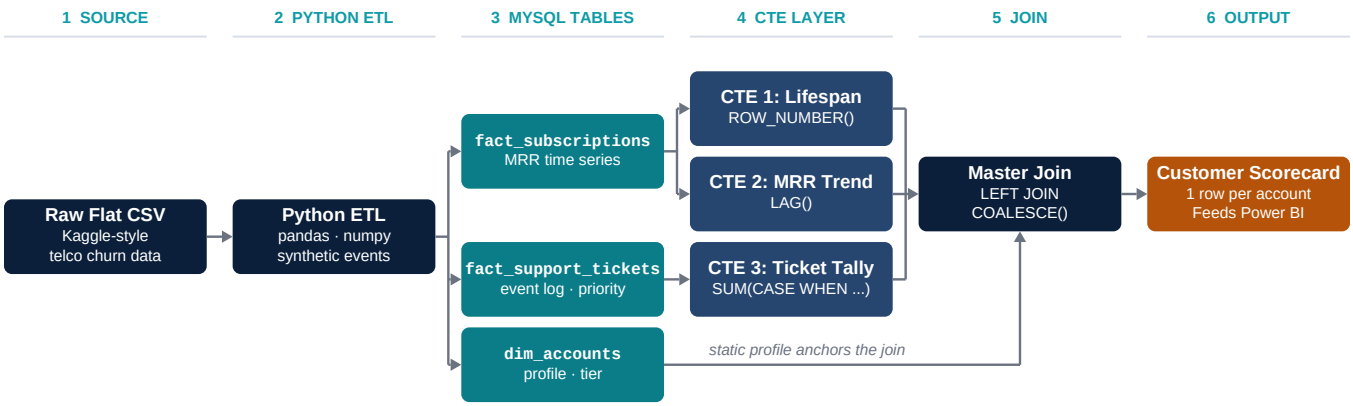
Table	Type	Purpose
`dim_accounts`	Dimension	Static customer profiles and tier segmentation; the anchor for every join
`fact_subscriptions`	Fact	Time-series log of Monthly Recurring Revenue (MRR) changes per account
`fact_support_tickets`	Fact	Event log of customer-service interactions, including priority level

THE ANALYTICAL ENGINE: MASTER SQL QUERY

A single, high-performance query validates the hypothesis end to end. Each technique was chosen to solve a specific analytical problem, not for show.

Technique	Key Constructs	Business / Engineering Value
Common Table Expressions	`WITH ... AS`	Modular, readable logic that stages each metric independently and keeps the query memory-efficient
Window Functions	`ROW_NUMBER()`	Computes each account's chronological lifespan, i.e. its active month, to place events inside the 7th to 11th month window
Window Functions	`LAG()`	Tracks month-over-month MRR fluctuations to surface revenue contraction ahead of cancellation
Conditional Aggregation ("Tally Mark" pattern)	`SUM(CASE WHEN priority = 'Escalated' THEN 1 ELSE 0 END)`	Compresses 12,500+ raw ticket rows into one row per customer, eliminating join fan-out and duplication
Advanced Joins	`LEFT JOIN, COALESCE()`	Stitches the dynamic CTEs back to static dim_accounts, preserving every customer and defaulting missing events to zero

PIPELINE TREE: FROM RAW CSV TO CHURN SCORECARD



FINDINGS

- **Hypothesis supported.** High-risk Tier 2 customers were successfully isolated from the wider base.
- **A clear warning signal.** Account cancellations were preceded by a spike in escalated, unresolved support tickets during months 7 to 11 of the subscription.
- **Actionable lead time.** The pattern appears before churn occurs, which makes it usable as an early-warning trigger rather than a post-mortem.
- **BI-ready output.** The final clean, one-row-per-customer dataset is prepared for Power BI dashboarding.

RECOMMENDED BUSINESS ACTIONS

- Route Tier 2 accounts with escalating tickets in months 7 to 11 to a dedicated retention or customer-success queue.
- Set an automated alert on escalated-ticket count combined with MRR decline (LAG-based) as a composite churn-risk flag.
- Prioritize first-contact resolution for premium customers during the identified high-risk window.

TECHNOLOGY STACK & SKILLS DEMONSTRATED

Python	**pandas**	**numpy**	**MySQL**
Star Schema Design	**ETL**	**Bulk Ingestion**	**CTEs**
Window Functions	**Conditional Aggregation**	**Power BI**	

Note: Event-level data (subscriptions and support tickets) was synthetically generated to emulate a live operational environment; the methodology, architecture and query design are directly transferable to production data.